

Modular & Hybrid Physical-Digital Twin for Optical-Mobile Networking

V. Brienza⁽¹⁾, L. Inglés^(1,2,3), C. Rattaro⁽¹⁾, L. Anet Neto^(2,3), M. Morvan^(2,3), A. Castro⁽¹⁾, L. Nuaymi⁽²⁾

⁽¹⁾ Universidad de la República, 565 Av. Julio Herrera y Reissig, 11300 Montevideo, Uruguay, valentin.brienza@fing.edu.uy

⁽²⁾ IMT Atlantique, 655 Av. du Technopole, 29280 Plouzané, France

⁽³⁾ Lab-STICC CNRS UMR 6285, Technopôle Brest-Iroise - CS 83818, 29238 Brest Cedex 3, France

Abstract We demonstrate a modular optical-mobile testbed combining commercial G-PON equipment, an OMNeT++/INET optical DWDM network simulator, and open-source 5G/O-RAN components. Its interchangeable physical, simulated, and emulated segments support researcher-defined fixed-mobile convergence experiments. ©2026 The Author(s)

Review

Fixed-mobile convergence is reshaping the role of optical access and metro networks, as mobile fronthaul, midhaul, and backhaul traffic increasingly rely on shared fixed optical infrastructures. This places additional stress on the optical segments that interconnect radio, core, and edge resources, where service-specific requirements such as capacity, latency, and reliability must be guaranteed despite variations in traffic load. These characteristics make mobile X-haul a relevant use case for optical-network experimentation, where application, radio, and optical access/metro metrics must be observed jointly to understand end-to-end service behavior [1–3].

Optical access and transport technologies provide the substrate for fixed-mobile convergence scenarios. At the access level, Passive Optical Networks (PONs) allow multiple Optical Network Units (ONUs) to share a common fiber through time-division multiplexing/multiple access over downstream/upstream wavelength pairs. Among them, Gigabit PON (G-PON) [4], one of the most widely deployed technologies in current fixed access infrastructures, organizes downstream transmission in continuous fixed 125 μ s frames broadcast by the Optical Line Terminal (OLT). In contrast, upstream transmission occurs in bursts scheduled via the Bandwidth Map (BWMap) and carried in the G-PON Transmission Convergence (GTC) header, with the same 125 μ s periodicity. This scheduling process is governed by Dynamic Bandwidth Allocation (DBA) mechanisms and Traffic Container (T-CONT) parameter combinations, which determine how upstream capacity is allocated and, in turn, affect latency, throughput, and fairness [5]. At the metro level, Elastic Optical Networks (EONs) provide flexible lightpath configuration through fine-grained spectrum allocation and adaptive resource assignment. Routing and Spectrum Assignment (RSA) mechanisms are therefore central to EON operation over meshed networks, since they deter-

mine the path and spectrum resources assigned to each demand under dynamic traffic conditions [6].

On the mobile side, the Open Radio Access Network (O-RAN) architecture introduces a disaggregated and programmable approach to Radio Access Network (RAN) design [7]. Its main goal is to replace closed, vertically integrated RAN deployments with open interfaces, virtualized functions, and intelligent control loops, while balancing RAN centralization with the bitrate and latency constraints of RAN interfaces. In this architecture, the RAN Intelligent Controller (RIC) enables external applications to monitor and optimize RAN behavior. At the same time, interfaces such as E2 connect the Near-Real-Time RIC with RAN nodes and expose selected measurements, configuration, and context information. These capabilities are particularly relevant for fixed-mobile convergence experiments, since RAN-level information such as per-user throughput, radio conditions, and resource allocation can be made available for correlation with application-level and optical-domain observations [8].

Novelty

Previous works have addressed wireless-optical integration, programmable compute-RAN-optical infrastructures, and multi-site 5G experimentation workflows [8–12]. However, they typically focus on large-scale facilities, orchestration frameworks, monitoring architectures, or individual network segments. Our work instead emphasizes a compact, modular, researcher-configurable platform combining physical, simulated, and emulated optical-mobile segments.

The novelty of the proposed solution lies in allowing these physical, simulated, and emulated segments to be assembled according to the research question under study. Instead of proposing a single monolithic testbed, the platform provides a set of interchangeable building blocks (Fig. 1). The first block comprises an SDN-enabled commercial G-PON testbed, providing a physical passive opti-

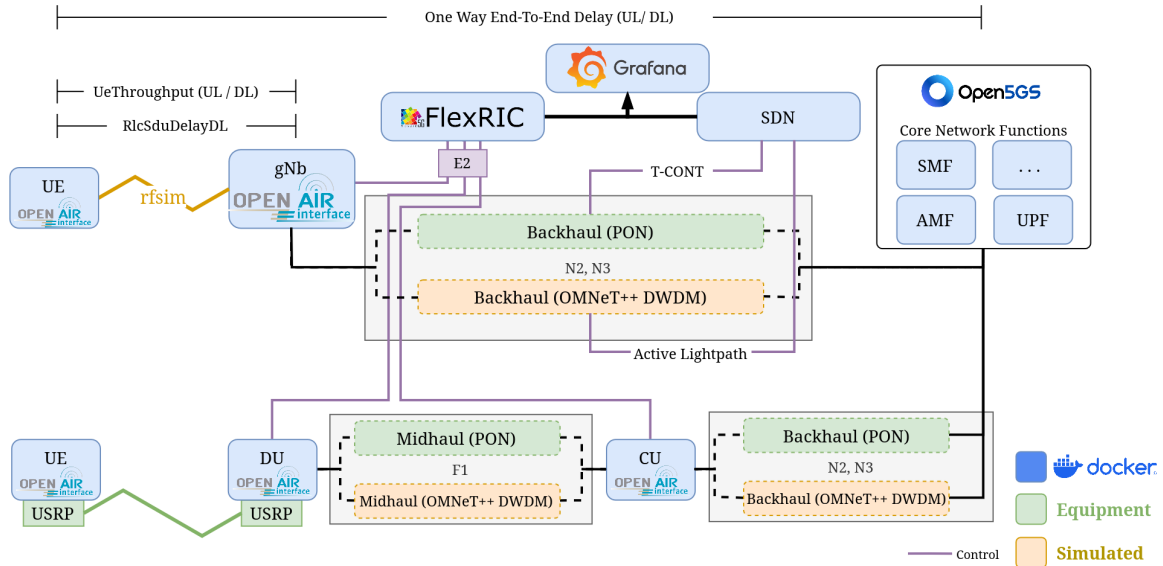


Fig. 1: Modular architecture of the proposed optical-mobile experimental platform.

cal access segment for evaluating 5G traffic over operational hardware [12]. The second block features an in-house OMNeT++/INET-based optical network simulator that implements the optical data and control planes for configurable EON scenarios. It incorporates a physical-layer impairment model, allows controlled variations in topology and link conditions, and supports the implementation and evaluation of different RSA mechanisms. The third block consists of an open-source 5G/O-RAN environment—integrating OAI [13], Open5GS [14], and FlexRIC [15]—capable of generating and processing realistic mobile traffic.

A distinctive aspect of the 5G/O-RAN block is its ability to operate under different deployment modes and architectural configurations. In a fully emulated configuration, the RF simulator enables end-to-end 5G experiments without dedicated radio hardware and allows basic channel impairments to be introduced in software. In a hardware-assisted configuration, Software Defined Radio (SDR) equipment can be used on the gNB side to connect physical user equipment (UE) and evaluate the impact of radio conditions on user traffic. The platform also supports both a monolithic gNB and a CU-DU split configuration. When the split is enabled, the F1 interface can be transported through the optical domain, extending the scope of the experiments from backhaul-only scenarios to midhaul-aware evaluations. Through O-RAN-compatible telemetry, particularly via the E2 interface, the platform exposes per-UE metrics, including downlink and uplink throughput, signal quality indicators, and radio resource allocation information. These measurements can be correlated with application-level performance and optical-domain observations, creating the basis for cross-domain analysis across the mobile and optical layers.

All 5G/O-RAN components are containerized

with Docker, simplifying topology changes and improving portability across hardware configurations. From the perspective of the 5G stack, the underlying transport segment only needs to provide IP connectivity between the corresponding nodes. This decoupling makes the mobile deployment agnostic to the specific access or metro technology being evaluated. As a result, the same 5G/O-RAN configuration can be tested on a commercial G-PON testbed, a simulated optical segment, or alternative transport technologies such as DWDM or emulated links. The platform functions as a real-time co-simulation environment, integrating the discrete-event simulator with emulated or physical components. Specifically, live traffic can be injected into the discrete simulator, enabling the optical domain to process real-world payloads alongside simulated network traffic in real-time. Conversely, the optical components can be stressed with realistic mobile traffic generated by the 5G stack.

To illustrate the platform's flexibility, we report one representative configuration. A full 5G/O-RAN network is connected to a commercial G-PON testbed acting as the backhaul. Fig. 2 shows an experiment where two UEs (connected to a gNB) upload and download at 10 Mbps. Initially, the G-PON uses a T-CONT with a fixed information rate of 16.5 Mbps, causing congestion and high one-way uplink delay. After changing the T-CONT to 30.0 Mbps, the second half of the figure shows that the UEs increase uplink throughput and greatly reduce uplink delay. The same 5G/O-RAN configuration can replace the commercial G-PON platform with other blocks, such as the OMNeT++/INET-based optical simulator, illustrating a researcher-configurable framework that assembles commercial equipment, simulation, and emulation for fixed-mobile convergence studies.



Fig. 2: Grafana dashboard showing key performance indicators for two UEs connected to a gNB over a G-PON backhaul.

ECOC Relevance

This work addresses the convergence of optical access/metro networks with 5G deployments, providing an experimental platform for unified autonomous control based on joint optical and radio telemetry. The platform includes a fully emulated DWDM co-simulation environment, representing a step toward a digital twin approach.

It is aligned with autonomous control, orchestration, and telemetry-driven management. Through the E2 and O1 interfaces, mobile telemetry (throughput, signal quality, and resource allocation) is exposed to the same control loop as optical-domain metrics, enabling cross-domain orchestration policies driven by live data rather than static configuration.

Demo Implementation

The demo runs either 5G/O-RAN network and OMNeT++/INET-simulated optical segments entirely on-site on a single server, or remotely in the G-PON testbed. Two complementary experiments will be presented.

The first experiment reproduces the configuration presented in the previous section, where the 5G/O-RAN network uses the physical G-PON laboratory as its backhaul. In this setup, attendees will be able to evaluate different mobile traffic loads and observe how variations in the T-CONT parameter affect end-to-end performance. Application-level measurements will be obtained using concurrent UDP iperf3 flows across multiple UEs, reporting per-UE throughput, jitter, and packet loss. At the RAN layer, FlexRIC xApps will collect per-UE KPM metrics, including downlink and uplink throughput, PDCP statistics, and resource allocations through the E2 interface. These measurements will allow attendees to observe the impact of G-PON upstream configuration on both application

and RAN-level performance.

The second experiment explores the interaction between radio mobility and optical network conditions in a CU-DU split deployment. Here, the optical segment is provided by the OMNeT++/INET-based simulation environment, where attendees can modify link bandwidth, propagation delay, and packet loss, and immediately observe their effect on the 5G network. The topology is shown in Fig. 3, and two dynamic events can be triggered. First, a UE performs an intra-CU handover between two DUs, each connected to the core through a different simulated lightpath, allowing attendees to observe how lightpath characteristics affect user experience before, during, and after the handover. Second, one DU is rerouted to a different lightpath while the UE session remains active, emulating dynamic optical reconfiguration. The transition is characterized at both the RAN layer, through FlexRIC KPM reports, and the application layer, through iperf3 measurements.

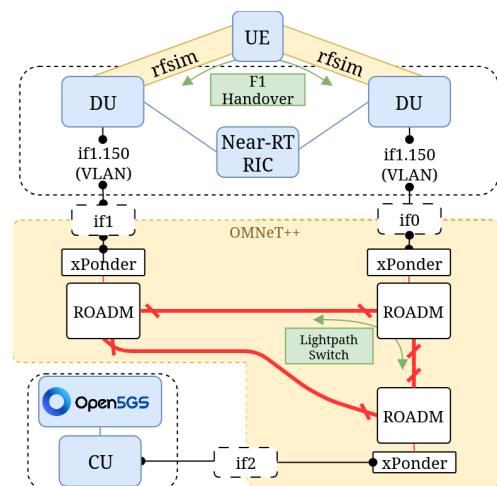


Fig. 3: Network architecture of the OMNeT++/INET demo scenario

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